

Artificial Intelligence

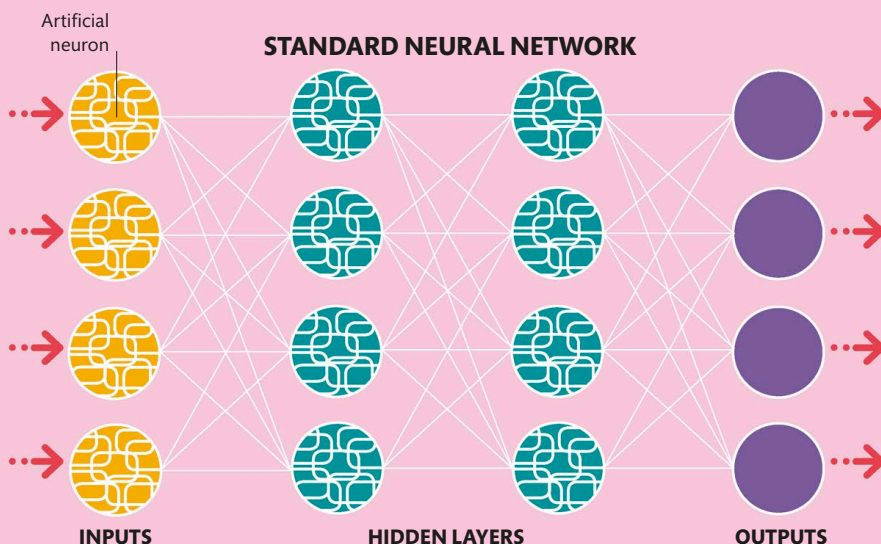
As computers become more sophisticated, the ultimate goal is to develop a machine that passes the Turing Test, in which a person in conversation with the machine cannot tell that they are not talking to another person.

Delivering dropout

Many electronic neural networks analyze and process in stages. In dropout, the probability is assessed that a particular item of information will or will not be useful. If it is not, it is removed.

Mimicking the brain

Computer programs called neural networks attempt to copy the way the brain works by using artificial neurons arranged in layers. Inspired by the way people learn, neural networks can adapt and change their responses over time (see right), a feature known as machine learning. To more closely replicate the human brain's highly adaptive, generalized intelligence, a more advanced approach involves querying, modifying, and deleting data, a technique called adaptive forgetting. For example, data that is little used further along a network, as shown by feedback through the system, can be trimmed or deleted. This is called dropout. Reducing this redundant data produces a system that is faster and more compact and responsive.



1 Input layer

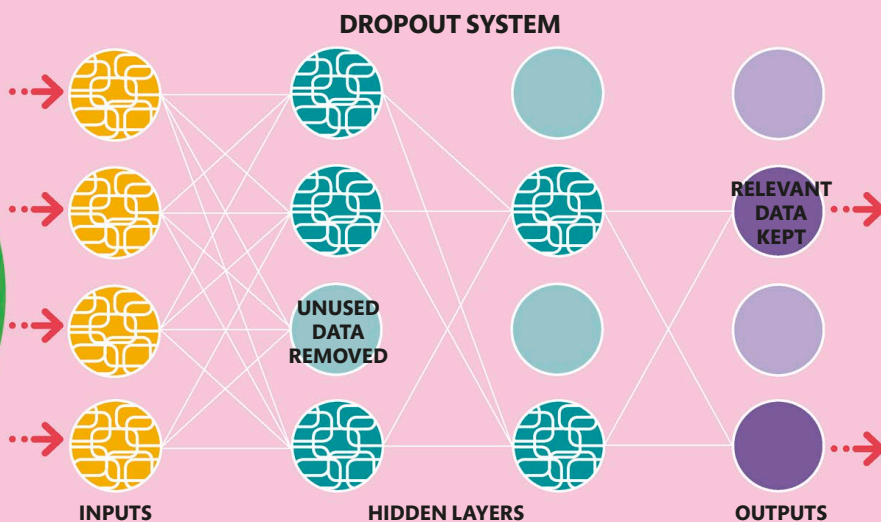
The network receives inputs in the form of numbers, or values. For example, in an image-recognition system, an input might be the brightness of an individual pixel in a digital image.

2 Hidden layers

The hidden layers process the data they receive from the input layer. Over time, the network "learns," modifying its results by applying different weights to the values.

3 Output layer

Once it has been processed, data passes to the output layer. In the image-recognition system, the output would be the application's "guess" for what the image shows.



WILL ROBOTS TAKE OVER THE WORLD?

An "AI takeover" sounds like science fiction, but it is hypothetically possible. A lot depends on friendly computers preventing self-evolving ones from advancing beyond humans.